ECE 4301
12/8/2025

Benchmarking Ascon

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Why Ascon and LWC Matters

- 18.5 billion IoT devices connected in 2024 [1]
- 32% use Wi-Fi, 24% use Bluetooth [1]
- Both need encryption [2][3]
  - However, encryption is computationally heavy
- Ascon/LWC close the performance-security gap [4]

1) https://iot-analytics.com/number-connected-iot-devices/
2) https://www.bluetooth.com/wp-content/uploads/Files/Specification/HTML/Core-54/out/en/low-energy-controller/low-energy-link-layer-security.html
3) https://documentation.meraki.com/Wireless/Design_and_Configure/Architecture_and_Best_Practices/Wireless_Fundamentals%3A_Encryption_and_Authentication
4) https://www.nist.gov/news-events/news/2023/02/nist-selects-lightweight-cryptography-algorithms-protect-small-devices

Original Project Idea

From the Project Vision

Project Vision

Studying the lightweight cryptographic primitives of ASCON, which uses sponge construction modeling using different generations of Raspberry Pi.

Raspberry Pi 5	Raspberry Pi 4b	Raspberry Pi 3b+	Raspberry Pi Zero W
ARM Cortex A-76	**ARM Cortex A-72**	**ARM Cortex A-53**	**ARM1176JZF-S**
Processor: BCM2712	Processor: BCM2711	Processor: BCM2837B0	Processor: BCM2835
2.4GHz	1.5GHz	1.4GHz	700MHz
Quad-core 64-bit	Quad-core 64-bit	Quad-core 64-bit	Single-core 32-bit
Up to 16 GB LPDDR4 SDRAM	Up to 8GB LPDDR4 SDRAM	1GB LPDDR2 SDRAM	512MB LPDDR2 SDRAM

Tools

We will use the tools from the NIST LWC Standardization Project that were previously used for benchmarking microcontrollers, and may supplement or replace with other tools if they are seen as more appropriate.
(https://github.com/usnistgov/Lightweight-Cryptography-Benchmarking)

Excel will be used for data analysis and visualization where the included Python scripts are not applicable.

Project roadmap

	Phase 1: Research	Phase 2: Collect Data	Phase 3: Interpret Results
Objectives	Research the Ascon algorithm and NIST benchmarking. Understand the differences between different Pi platforms. Consider any additional tests or feasible platform test additions.	Perform benchmarking. Collect enough data to do meaningful processing later.	Process results and report findings.
Activities	→ Find and understand papers on Ascon and review literature from other benchmarks/reports → Be specific and add as many as you need	→ Use NIST benchmark tools to benchmark on our available platforms. → Collect detailed data and demos.	→ Analyze what our results mean and feasibility of using Ascon in place of existing algorithms. → Organize all results in a readable and interesting way.
Deliverables	→ Report drafted section: understanding/evaluating Ascon → Goals for benchmark data (what equations to use, etc.)	→ Videos and screenshots of experimental procedure → Raw CSV files with results	→ Final report with analysis of results and Ascon architecture → Graphs and plots from data → Github repo with raw data, demos, report

All roles will be shared based on need, but certain members will lead:

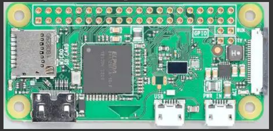
Research: Mia
Demo: John
Benchmarking/Data: Kelly
Documentation: Marcelo and Garima

The actual project

From our work

Hardware

Raspberry Pi Zero W

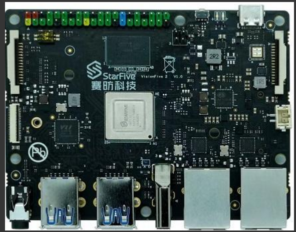


ARM1176JZF-S
Processor: BCM2835
700MHz
Single-core 32-bit
512MB LPDDR2 SDRAM

Raspberry Pi 5

ARM Cortex A-76
Processor: BCM2712
2.4GHz
Quad-core 64-bit
Up to 16 GB LPDDR4 SDRA

VisionFive 2

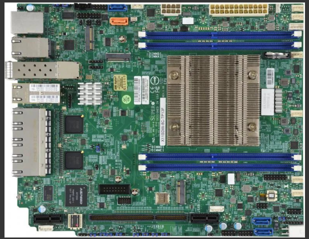


RISC-V U74
Processor: StarFive JH7110
1.5GHz
Quad-core 64-bit
Up to 8 GB LPDDR4 SDRAM

Qotom Mini PC

Intel Silvermont (Bay Trail x86-64)
Processor: Intel Bay Trail Celeron SoC
1.83 GHz to 2.66 GHz
Quad-core 64-bit
Up to 8 GB DDR3L

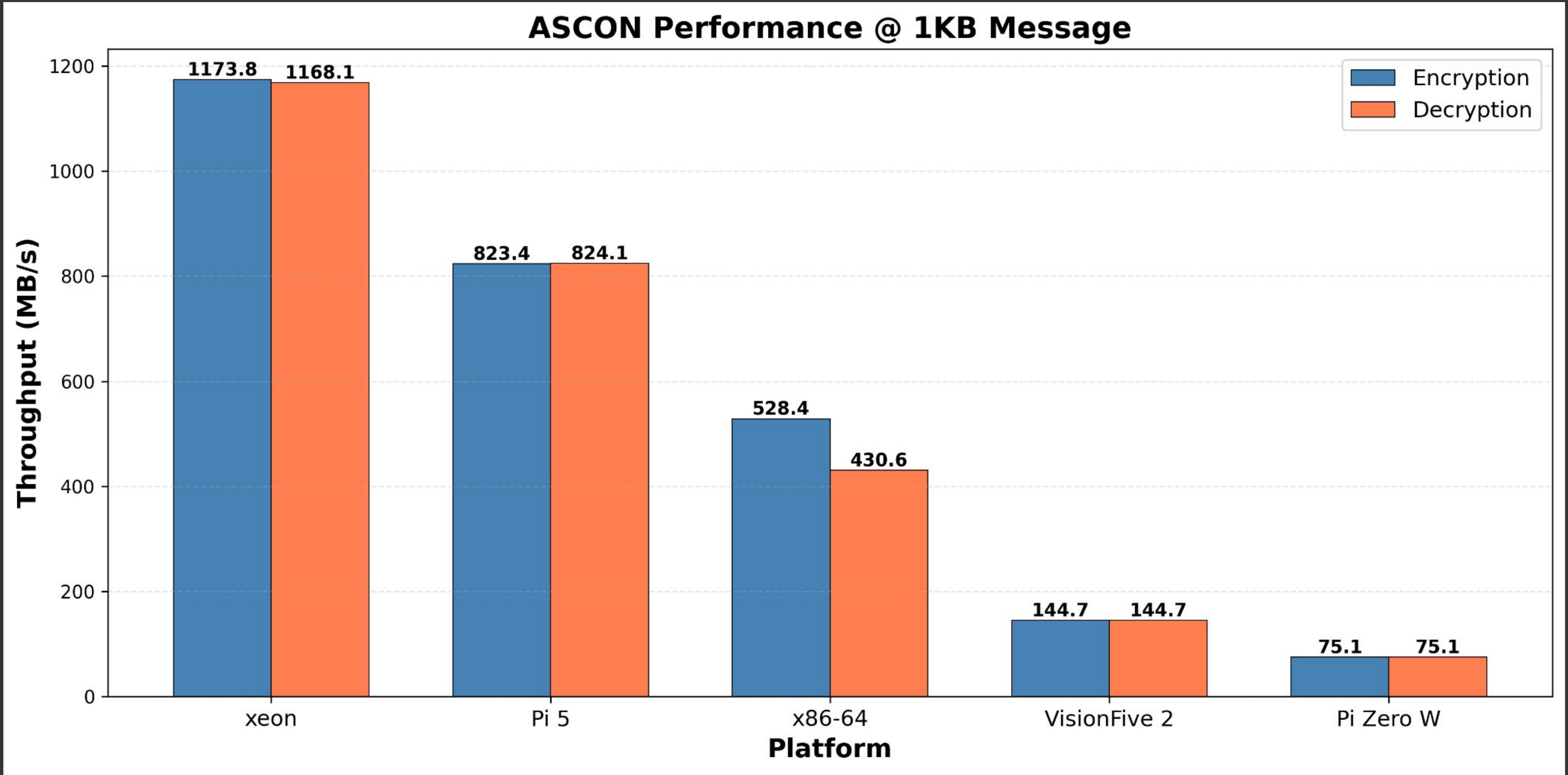
Supermicro X11SDW-8C-TP13F



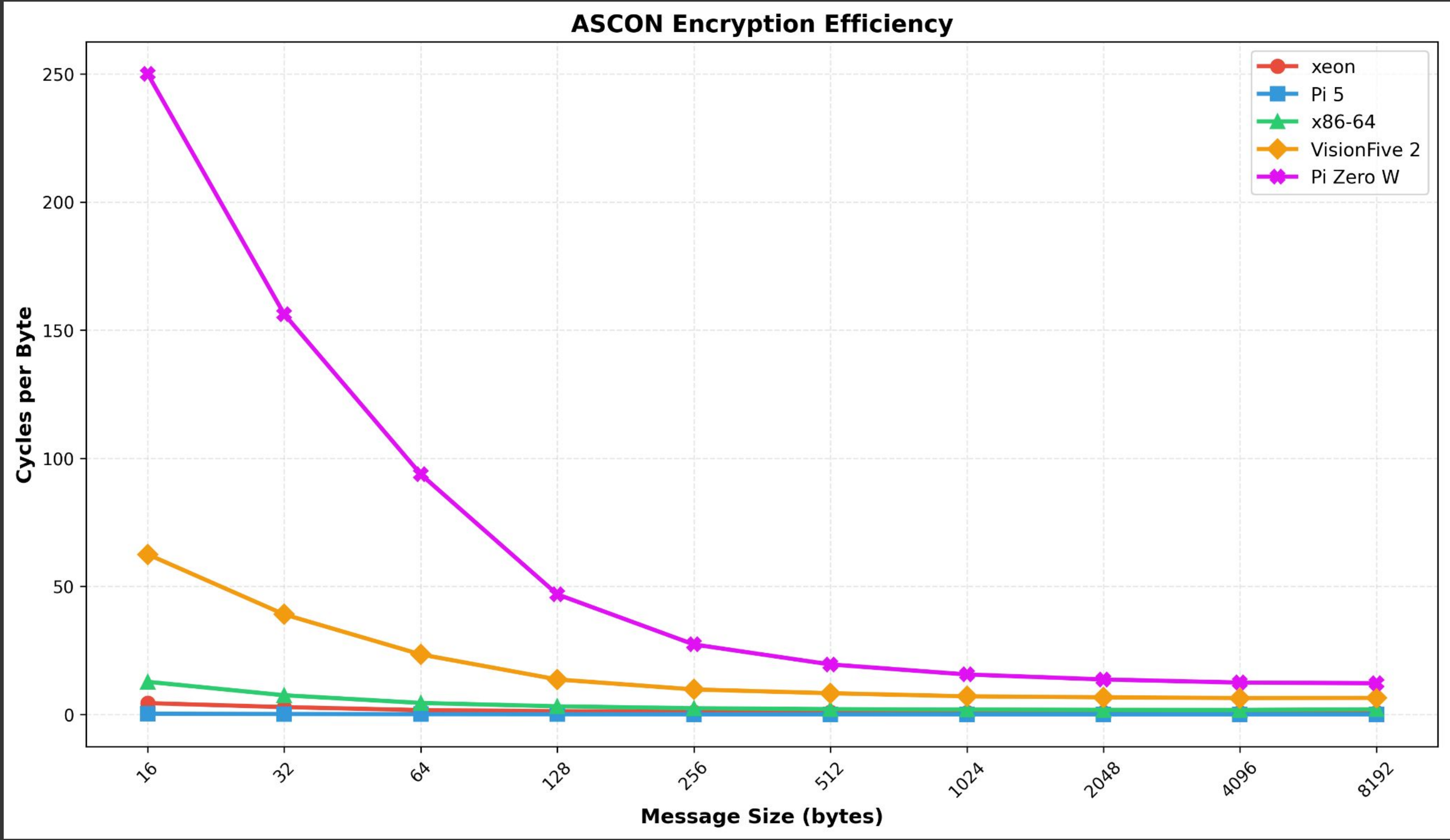
Intel Skylake-D (x86-64)
Processor: Intel Xeon D-2146NT SoC
2.3 GHz to 3.0 GHz
Up to 512 GB ECC DDR4-2133 LRDIMM

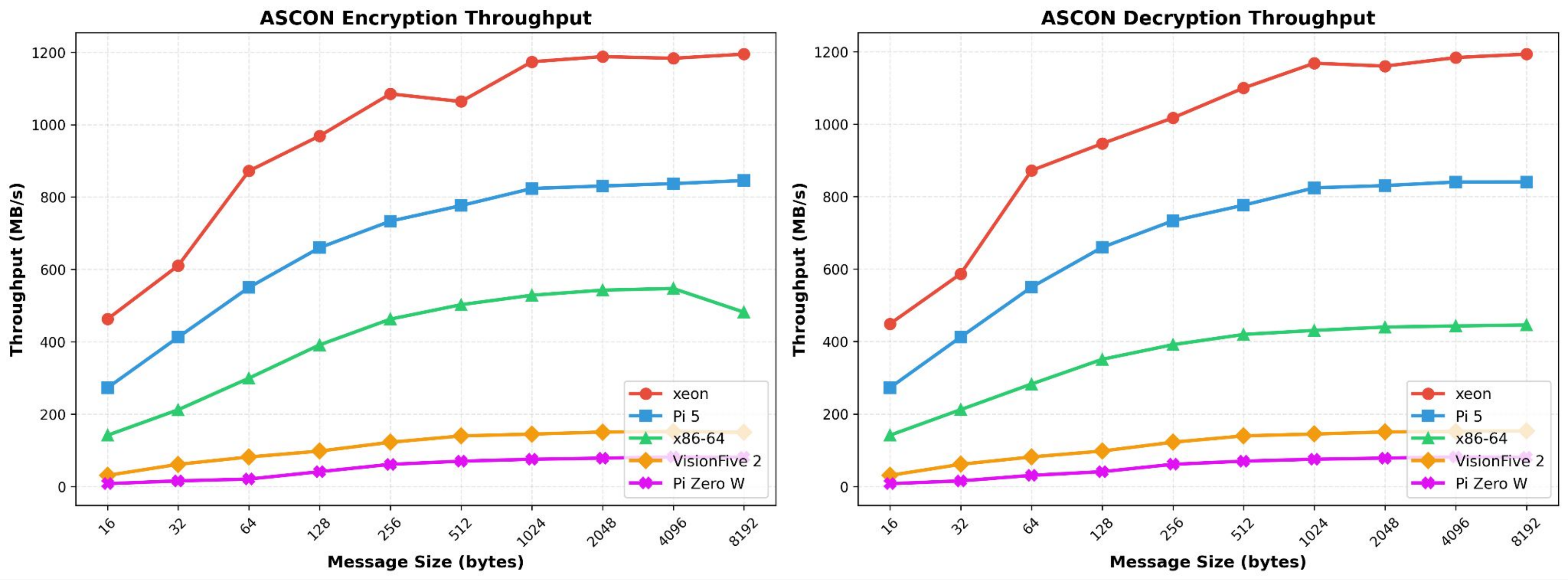
Studied the lightweight cryptographic primitives of ASCON, which uses sponge construction modeling using different three different architectures, ARM, RISC-V and x86

How to run the project



ASCON Performance @ 1KB Message
Throughput (MB/s)
1200
1000
800
600
400
200
0
1173.8
1168.1
823.4
824.1
528.4
430.6
144.7
144.7
75.1
75.1
Encryption
Decryption
xeon
Pi 5
x86-64
VisionFive 2
Pi Zero W
Platform


ASCON Encryption Efficiency
Cycles per Byte
Message Size (bytes)
xeon
Pi 5
x86-64
VisionFive 2
Pi Zero W
0
50
100
150
200
250
16
32
64
128
256
512
1024
2048
4096
8192



ASCON Encryption Throughput
Throughput (MB/s)
0
200
400
600
800
1000
1200
16
32
64
128
256
512
1024
2048
4096
8192
Message Size (bytes)
xeon
Pi 5
x86-64
VisionFive 2
Pi Zero W
ASCON Decryption Throughput
Throughput (MB/s)
0
200
400
600
800
1000
1200
16
32
64
128
256
512
1024
2048
4096
8192
Message Size (bytes)
xeon
Pi 5
x86-64
VisionFive 2
Pi Zero W

Any questions?
Ask away!

Thank you